EASTERN INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY

Practice Assignment – Quarter 1, 2026- 2027

Course Name: Coding Practice
Course Code: CSE 422
Student's Full Name: Thai Phan Thanh Sang
Student ID: 2231200114

LAB 04

GitHub Link: https://github.com/thaiphanthanhsang/CSE422_CodingPractice.git

1. Problem Statement and Requirements

1.1 Functional requirements

Area	Required behavior
Books	Add a title, author, category and quantity; allocate a unique book ID.
Search	Find books by title or category using case-insensitive keyword matching.
Readers	Maintain readers with unique IDs; list and add reader records.
Borrow	Permit a loan only when a copy/license is available and the reader has fewer than three active loans.
Return	Close a valid active loan and increment available quantity.
Reports	Show each reader, borrowed title and status (Borrowing or Returned).

1.2 Technical requirements

- Apply all five SOLID principles using cohesive classes and focused interfaces.
- Separate book management, reader management, borrowing services and reporting.
- Support different book types with inheritance and an extensible factory abstraction.
- Provide sample data and automated/manual testing scenarios.

2. Domain Model and UML

The core domain is built around three entities: Book, Reader and Loan. Book is abstract; PrintedBook, EBook and Magazine are substitutable subtypes. Each borrowing operation creates a Loan that links one Reader and one Book.

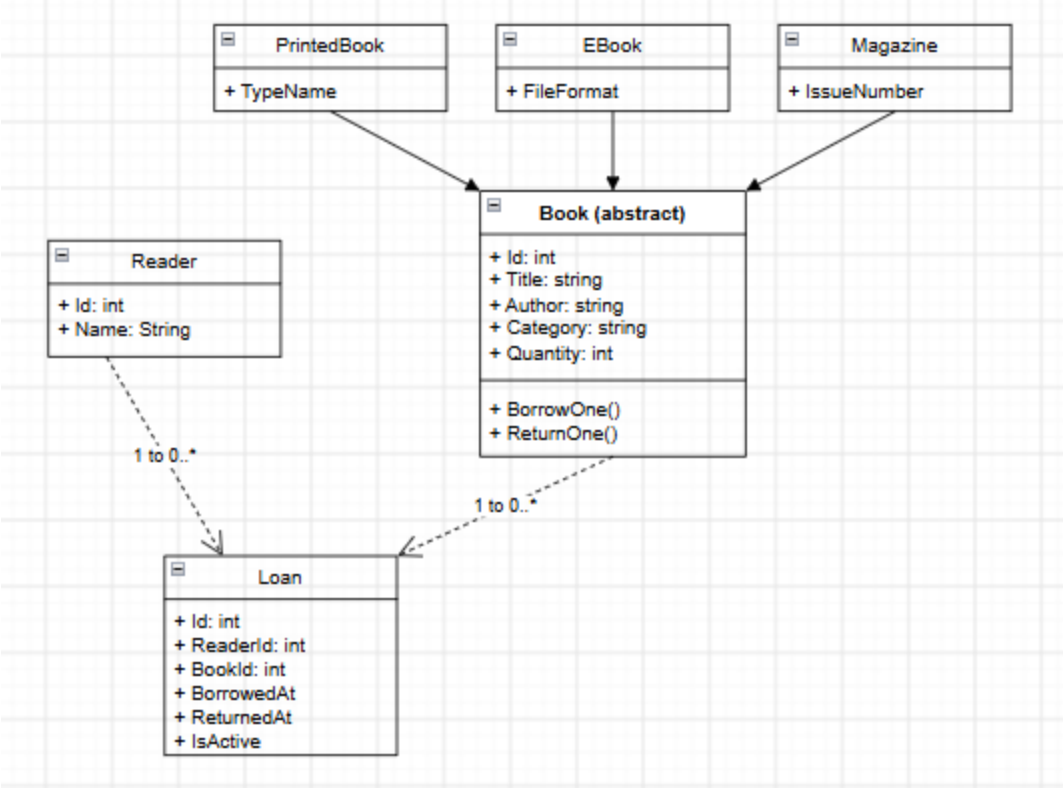


Figure 1. Domain UML: inheritance, book subtypes and loan relationships

Domain invariants

- IDs must be positive and unique in their respective repositories.
- Book title, author and category cannot be blank; quantity cannot be negative.
- Loan keeps BorrowedAt, optional ReturnedAt and derived IsActive status.

3. Architecture and Interface Design

The program follows a layered organization. The console menu calls interfaces, the service layer enforces use cases, repositories encapsulate storage, and domain models protect internal state. Program.cs is the composition root that creates concrete objects and injects them into services.

Main interfaces and implementations

Interface	Concrete class / role
IBookService	BookService: add, list, retrieve and search titles/categories
IReaderService	ReaderService: add, list and retrieve readers
ILendingService	LendingService: borrow/return operations
IReportService	ReportService: reader borrowing history
IBookRepository	InMemoryBookRepository: book data access
IReaderRepository	InMemoryReaderRepository: reader data access
ILoanRepository	InMemoryLoanRepository: loan data access

IBorrowingPolicy	MaxThreeBooksPolicy: loan-count and duplicate-loan rules
IBookFactory	PrintedBookFactory / EBookFactory / MagazineFactory
IIdGenerator	SequentialIdGenerator: unique ID allocation

4. **Application of SOLID Principles**

S - Single Responsibility Principle

BookService handles the book catalogue; ReaderService manages readers; LendingService manages borrowing and returns; ReportService produces report text; MaxThreeBooksPolicy enforces borrowing restrictions. Each has one main reason to change.

O - Open/Closed Principle

A new media type such as AudioBook can subclass Book and receive its own IBookFactory implementation. The book catalogue, lending and reporting services operate on Book and do not need changes. A new policy can be introduced by implementing IBorrowingPolicy.

L - Liskov Substitution Principle

PrintedBook, EBook and Magazine share the invariant-preserving BorrowOne/ReturnOne behavior inherited from Book. All three can be used through the Book abstraction wherever a lendable item is expected.

I - Interface Segregation Principle

Separate service interfaces (IBookService, IReaderService, ILendingService, IReportService) and repository interfaces keep contracts task-specific. Classes are not forced to implement unrelated operations.

D - Dependency Inversion Principle

LendingService receives IBookRepository, IReaderRepository, ILoanRepository, IBorrowingPolicy and IIdGenerator through constructor injection. InMemory repositories are created only in Program.cs, so storage may later be replaced.

5. **Implementation and Sample Data**

LibraryManagementSOLID/

Models/	Book, PrintedBook, EBook, Magazine, Reader, Loan
Interfaces/	Service, repository, factory, policy contracts
Repositories/	In-memory data access
Services/	BookService, ReaderService, LendingService, ReportService
Policies/	MaxThreeBooksPolicy
Factories/	Three book factories
Infrastructure/	SequentialIdGenerator
Data/	SampleDataSeeder
Tests/	SmokeTests
Docs/	UML diagrams and design notes
Program.cs	Application entry point and dependency wiring

6. **Demo**

```
========== LIBRARY MANAGEMENT (SOLID) ==========
1. List books                    2. Add a book
3. Search by title               4. Search by category
5. List readers                  6. Add a reader
7. Borrow a book                 8. Return a book
9. Borrowing report              0. Exit
Choose: 1
```

```
Choose: 1
ID | Type         | Title                      | Author              | Category       | Stock
-------------------------------------------------------------------------------------------
 1 | Printed Book | Clean Code                 | Robert C. Martin    | Programming    |     3
 2 | Printed Book | Design Patterns            | GoF                 | Programming    |     2
 3 | Printed Book | The Pragmatic Programmer   | Hunt & Thomas       | Programming    |     1
 4 | Printed Book | Database Systems           | Ramez Elmasri       | Database       |     0
 5 | eBook        | C# Digital Handbook        | Learning Team       | Programming    |     2
 6 | Magazine     | Science Monthly            | Editorial Board     | Science        |     2
```

```
Choose: 2
Available types:
  1: Printed Book
  2: eBook (quantity = concurrent licenses)
  3: Magazine
Type key: 3
Title: Today
Author: A
Category: life
Quantity: 10
Book added. New ID: 7
```

```
Choose: 1
ID | Type         | Title                      | Author              | Category       | Stock
-------------------------------------------------------------------------------------------
 1 | Printed Book | Clean Code                 | Robert C. Martin    | Programming    |     3
 2 | Printed Book | Design Patterns            | GoF                 | Programming    |     2
 3 | Printed Book | The Pragmatic Programmer   | Hunt & Thomas       | Programming    |     1
 4 | Printed Book | Database Systems           | Ramez Elmasri       | Database       |     0
 5 | eBook        | C# Digital Handbook        | Learning Team       | Programming    |     2
 6 | Magazine     | Science Monthly            | Editorial Board     | Science        |     2
 7 | Magazine     | Today                      | A                   | life           |    10
```

```
Choose: 3
Title keyword: Today
ID | Type         | Title                      | Author              | Category       | Stock
-------------------------------------------------------------------------------------------
 7 | Magazine     | Today                      | A                   | life           |    10
```

```
Choose: 4
Category keyword: Programming
ID | Type         | Title                      | Author              | Category       | Stock
-------------------------------------------------------------------------------------------
 1 | Printed Book | Clean Code                 | Robert C. Martin    | Programming    |     3
 2 | Printed Book | Design Patterns            | GoF                 | Programming    |     2
 3 | Printed Book | The Pragmatic Programmer   | Hunt & Thomas       | Programming    |     1
 5 | eBook        | C# Digital Handbook        | Learning Team       | Programming    |     2
```

```
Choose: 5
Reader #1: Nguyen Van An
Reader #2: Tran Thi Binh
Reader #3: Le Minh Chau
```

```
Choose: 6
Reader name: Thanh Sang
Reader added. New ID: 4
```

```
Choose: 5
Reader #1: Nguyen Van An
Reader #2: Tran Thi Binh
Reader #3: Le Minh Chau
Reader #4: Thanh Sang
```

```
Choose: 7
Reader ID: 4
Book ID: 7
Borrowed successfully. Loan ID: 1
```

```
Choose: 8
Reader ID: 4
Book ID: 7
Book returned successfully.
```

```
Choose: 9
========== BORROWING REPORT ==========
Lists both active loans and returned-book history.
Reader #1: Nguyen Van An | Active: 0
  (No loans)
Reader #2: Tran Thi Binh | Active: 0
  (No loans)
Reader #3: Le Minh Chau | Active: 0
  (No loans)
Reader #4: Thanh Sang | Active: 0
  Loan #1 | Book #7: Today | Returned | Borrowed: 2026-11-02 11:16 UTC
======================================
```

```
Choose: 0
Goodbye!

D:\Sang\CSE422\Lab4\Library_Management_SOLID_Source\LibraryManagementSOLID\bin\Debug\net10.0\LibraryManagementSOLID.exe
(process 16072) exited with code 0 (0x0).
Press any key to close this window . . .
```